

Wyatt Walsh

SENIOR AI/ML ENGINEER

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Senior AI/ML engineer shipping production agentic AI, LLM document intelligence, retrieval workflows, AI guardrails, data platforms, and fintech tooling.

EXPERIENCE

Senior AI/ML Engineer

Frame Payments | Remote | Oct 2025 - Jan 2026

Delivered production agentic AI document intelligence, human-in-the-loop review, and multimodal compliance screening for high risk merchant onboarding.

- Deployed production LLM document intelligence with PydanticAI, Amazon Bedrock AgentCore, and Claude; added human-in-the-loop browser feedback for confidential uploads and reduced analyst follow-ups/resubmissions ~90% by blocking miscategorized documents before review.
- Built multimodal restricted-item compliance screening by combining product images and structured metadata; translated banking-partner policy requirements into automated risk checks for merchant onboarding across tens of merchants and multi-location catalogs.
- Built a NotebookLM-based technical knowledge hub spanning vendor/web documentation and internal monorepo docs; automated corpus refresh with a Colab pipeline that re-ingests the latest sources to keep retrieval accurate and up to date.
- Drove internal AI adoption with Notion MCP playbooks, high-signal prompt patterns, and weekly office hours reaching ~15% of the company; standardized PII/confidential-document safeguards and auditability requirements.

Associate III, Senior Software Engineer - Data Integrations

JPMorgan Chase & Co. | New York, NY | May 2023 - Apr 2025

Built enterprise data pipelines, risk analytics, and data-quality automation for credit trading while scaling secure internal AI developer tooling.

- Pioneered firm-wide rollout of secure generative AI developer tools (LLMs, GitHub Copilot, Codeium) as one of the first 500 engineers, shaping scalable deployment and leading workshops for technical and non-technical teams.
- Devised ML-driven anomaly detection for credit trading and risk analytics, reducing false positives by 25% and manual review time for risk teams.
- Developed a resilient Python wrapper for KDB+ with CI-style tests and optimized Q functions, democratizing data access for 5,000+ users.
- Engineered a cloud-function-driven asset-fee data sink, consolidating 10+ sources into a centralized database, cutting manual maintenance by 90% and providing on-demand access for traders through a Python SDK, REST API, and interactive dashboard.

Senior Software Engineer

Stealth Web3 Startup | Remote | Mar 2022 - Jul 2022

- Architected a scalable collateralized lending app with React, Next.js, TypeScript, MobX, and ethers.js for secure wallet and smart-contract workflows.
- Implemented CI/CD with GitHub Actions, Husky, Cypress, and Jest, halving release cycles, raising test coverage to 90%, and mentoring a front-end intern.
- Optimized front-end performance through code splitting, lazy loading, and asset caching, reducing initial load times by 40% and JavaScript bundle size by 30%.

Founder, Chief Technology Officer

SandLabs | Remote | Jun 2021 - Mar 2022

- Launched a Next.js/TailwindCSS public site with top-tier Lighthouse scores and secured \$75K in competitive ecosystem grant funding.
- Built Python pipelines that normalized a 1M+ Reddit posts/comments dataset into a reusable analytics data product for external users.

Demand Forecasting Extern

Gap Inc. | San Francisco, CA | Jan 2020 - May 2020

- Applied time-series decomposition and bias correction techniques to demand forecasts, boosting accuracy by up to 25% and minimizing surplus inventory costs.
- Owned the data engineering layer by architecting scalable MySQL and Apache Spark pipelines across 5+ years of sales data, improving data processing throughput by 95% and standardizing insights across forecasting teams.

SKILLS

MACHINE LEARNING AND GENERATIVE AI: scikit-learn, Hugging Face Transformers, OpenAI API, LangChain, LangGraph, Model Context Protocol (MCP), Retrieval Augmented Generation (RAG), Prompt Engineering, PydanticAI, Amazon Bedrock, Claude, GitHub Copilot, Document AI, AI Guardrails

DATA ENGINEERING AND ANALYTICS: Pandas, NumPy, SQLAlchemy, Pydantic, Pandera, Apache Spark, AWS Glue, KDB+, PostgreSQL, DynamoDB, ETL, DuckDB, Polars, Parquet

PROGRAMMING & SCRIPTING LANGUAGES: Python, SQL, TypeScript, JavaScript, Go, Rust, Bash

CLOUD INFRASTRUCTURE AND DEVOPS: AWS, Docker, Kubernetes, Terraform, GitHub Actions, AWS Lambda, Google Cloud Functions, ECS, EKS

PROJECTS

AI Agent Harness Configs | github.com/wyattowalsh/agents

Jan 2026 - Present | Python, TypeScript, MCP, FastMCP

Open-source agent tooling catalog and docs portal spanning 40+ reusable skills, shared instructions, MCP-ready configs, and cross-runtime workflows for Claude Code, Codex, Copilot, Cursor, and related agents.

- Built 40+ instruction and skill packs plus harness configs across Claude Code, Codex, GitHub Copilot, Cursor, and related IDE agents.
- Shipped generated docs, install flows, and portable tool and MCP presets for code review, security, infrastructure, research, and workflow automation.

NBA Basketball Database | github.com/wyattowalsh/nbadb

Apr 2021 - Present | Python, Pandas, Polars, SQLAlchemy, DuckDB, Pandera

Open-source star-schema NBA analytics warehouse spanning 65K+ games and 13M+ play-by-play events, with validated ETL, docs, CLI, DuckDB/SQLite/Parquet/CSV exports, and Kaggle distribution.

- Implemented Pandera-validated NBA API ETL with daily warehouse updates plus DuckDB, SQLite, Parquet, and CSV exports.
- Built docs, schema references, release workflows, and Kaggle distribution for a 180+ table public warehouse spanning 130+ stats.nba.com endpoint classes.

ProxyWhirl | github.com/wyattowalsh/proxywhirl

May 2025 - Present | Python, SQLite, PyPI

Released rotating proxy toolkit with 9 selection strategies, 100+ auto-validated sources, and self-healing health checks for resilient scraping and data-collection workflows.

- Implemented 9 proxy rotation strategies with configurable fallback chains, per-source scoring, and performance-based selection.
- Built self-healing validation for 100+ proxy sources with circuit breakers, auto-ejection, and recovery promotion.

AI/ML Web Feeds | github.com/wyattowalsh/ai-web-feeds

Oct 2025 - Present | Python, Pydantic, SQLAlchemy, YAML, Fumadocs, Vercel

Built a curated AI and ML feed platform with typed schemas, HTTP/RSS validation, contributor submission workflows, OPML exports, and docs-backed browse/download surfaces.

- Defined YAML feed schemas with topic and source-type validation across blogs, papers, and newsletters for reliable feed quality.
- Shipped helper scripts, CI checks, scheduled validation jobs, and live docs for reliable community feed review.

MCP-Crawl4AI | github.com/wyattowalsh/mcp-crawl4ai

Apr 2025 - Present | Python, FastMCP, Crawl4AI, Playwright, Docker

Released FastMCP server wrapping Crawl4AI so AI agents can crawl and extract web content through four focused tools, agent-friendly outputs, tagged releases, and documented stdio/HTTP integration paths.

- Built a FastMCP v3 server with managed headless Chromium, session-aware crawling, and automatic Playwright setup using two runtime dependencies.
- Exposed stdio/HTTP tools, resources, and prompt templates with docs, tests, and release-tagged MCP install flows.

Personal Website: w4w.dev | github.com/wyattowalsh/personal-website

Nov 2018 - Present | Next.js, React, TypeScript, Tailwind CSS, MDX

Long-running personal web app and MDX publishing system built with Next.js, combining a polished public-facing frontend with a richly tooled technical blog, strong SEO, and durable content workflows.

- Maintained the public site across multiple generations, covering responsive landing pages, themes, motion, accessibility, and MDX publishing.
- Built custom MDX publishing with search, metadata, math rendering, feeds, tags, and integrated authoring tools.

EDUCATION & CERTIFICATIONS

Bachelor of Science

University of California, Berkeley - College of Engineering, Industrial Engineering and Operations Research (IEOR)

Aug 2014 - Dec 2020

High School Diploma

The Hotchkiss School

Aug 2012 - May 2014

AWS Certified Cloud Practitioner

Amazon Web Services (AWS) | Nov 2024

IBM Data Science Professional Certificate

IBM | Feb 2021